
The concepts of bioenergetics

Course for first-year medical students

Prepared by: [Dr. A. BOUFAMMA](#)

Educational objectives:

- 1- Explain the principles of thermodynamics
- 2- Define Gibbs free energy as a criterion for the spontaneity of a reaction
- 3- Differentiate between exergonic and endergonic reactions, as well as their role in the coupling of biochemical processes.
- 4- List energy-rich compounds.

Course outline:

- I. Introduction
- II. Prerequisites: definitions
- III. The Concept of Free Energy
- IV. Additivity of ΔG : reaction coupling mechanism
- V. Free activation energy $\Delta G^\#$
- VI. The concept of "energy-rich" compounds or high energy potential

I. INTRODUCTION

Living beings are the seat of a continuous flow of energy. Through photosynthesis, plants convert solar energy into chemical energy stored in organic matter such as carbohydrates. Animals then draw on this energy by metabolizing organic matter to power vital mechanisms such as biomolecule synthesis, concentration gradient maintenance, and muscle contraction. These processes ultimately transform energy into heat, which is dissipated into the environment. A significant portion of cellular biochemical apparatus must therefore be devoted to the acquisition and use of energy.

Knowledge of thermodynamics, the discipline that studies the principles governing energy exchanges, allows us to determine whether a process is possible or not. Thermodynamics is therefore essential for understanding:

- Why do macromolecules fold into their native conformations?
- How metabolic pathways are designed,
- Why do molecules cross biological membranes, etc.?

In biochemistry, thermodynamics is particularly useful for defining the conditions that allow processes to occur spontaneously (on their own).

II. PREREQUISITES: DEFINITIONS

Thermodynamics: is the quantitative description of energy exchanges between work and heat.

Bioenergetics: the field of biochemistry that studies the transformation and use of energy by living cells.

System: A system is any portion of space (the universe) that contains the matter being studied. Everything surrounding this system constitutes its environment (external environment).

Autotrophs: organisms capable of producing chemical energy from inorganic elements; this chemical energy is present in the form of complex organic molecules. There are two types:

- Phototrophic organisms, which are capable of using light energy and converting this energy into chemical steps.
- Chemotrophic organisms that use only chemical energy recovered during the oxidation of inorganic compounds to produce organic compounds.

Heterotrophs: organisms that are unable to synthesize their own constituents from mineral elements. They obtain their free energy by coupling their metabolism to the breakdown of complex organic molecules in their environment.

Energy: The ability of a system to perform work. It can take various forms: heat, electricity, chemical, mechanical, or light.

Another form of energy: Heat ; $\text{Energy} = \text{Work} + \text{Heat}$

Energy is expressed in joules (or calories): $1 \text{ cal} = 4.18 \text{ J}$; 1 cal: The amount of energy required to raise the temperature of 1g of water by 1 degree Celsius.

Entropy (S): A quantitative expression of randomness and disorder in a system; it is the degree of freedom between the components of a system or universe. Any spontaneous process must cause an increase in the entropy of the universe. It is a form of degraded energy that cannot be used for work.

Enthalpy H: This corresponds to the total energy (potential, heat) contained in a compound or system. It evaluates the stability and binding energy of molecules.

Depending on the changes in enthalpy (ΔH) between the initial and final states of a reaction, we say:

- **Exothermic:** When a chemical reaction releases heat, the thermal content of the products is lower than that of the reactants, and ΔH , by convention, has a negative value.
- **Endothermic:** describes reaction systems that absorb heat from their environment and have positive ΔH values.

III. THE CONCEPT OF FREE ENERGY

Every cell is the site of thousands of biochemical reactions involving the transfer of matter and energy. This set of reactions is called metabolism. The reactions form a network of highly branched pathways along which molecules, called metabolites, are transformed.

The cell constantly:

- Capture energy from the external environment (e.g., light energy).
- Transferring part of this energy to the external environment in the form of heat
- Transforming the "rest" of this energy into cellular work.

These energy movements are governed by the laws of thermodynamics.

1/ First law of thermodynamics: law of conservation of energy (Lavoisier 1773)

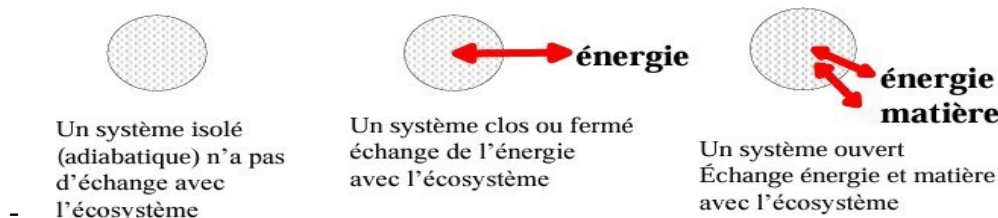
According to this principle, the total energy of a system and its surrounding environment (the universe) remains constant. Energy is neither created nor lost, but only transformed.

(Nothing is lost, nothing is created, everything is transformed)

$$E_{\text{system}} + E_{\text{environment}} = E_{\text{universe}} = \text{Constant}$$

A system can be:

- Open: exchange of matter and energy with the external environment, e.g., a cell;
- Closed: exchange of energy only with the external environment;
 - Isolated: no exchange with the external environment. E.g.: the universe.
 -
 -



The mathematical expression of the first law is :

$$\Delta E = EB - EA = Q - W \dots (1)$$

where, ΔE = change in internal energy

ΔEA = energy of a system at the start of a process

ΔEB = energy of a system at the end of a process

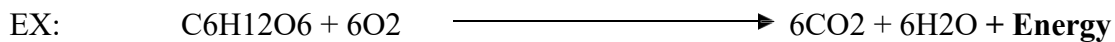
Q = heat absorbed by the system

W = work done by the system.

A noteworthy point about equation 1 is that the change in energy of a system depends only on the initial and the final stages and not on the path of transformation.

At the cellular level:

Energy production is represented in biochemistry by the oxidation or catabolism of biomolecules.



Some of the energy produced by the oxidation of metabolic fuels is released into the external environment in the form of heat, while some is stored in the form of ATP.

2 / Second law of thermodynamics: increase in entropy

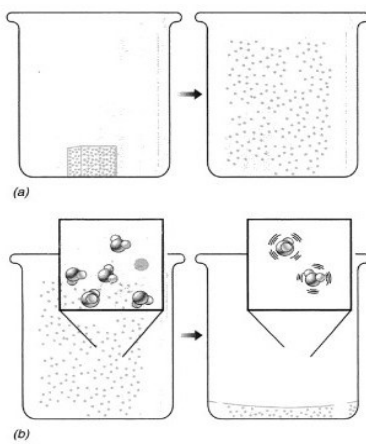
According to the second law of thermodynamics:

- The entropy of the universe constantly increases in any reaction.
- The total entropy of a system can decrease as long as the entropy of the environment increases by a higher value.

$$\Delta S_{\text{system}} + \Delta S_{\text{environment}} = \Delta S_{\text{universe}} > 0$$

Entropy (S) represents the degree of disorder or chaos (degree of freedom). It increases if a process occurs spontaneously and reaches its maximum in a system approaching equilibrium.

In a cellular system, the increase in entropy (disorder) is compensated for by processes that create order in the environment.



(a) A sugar cube contains sucrose molecules arranged in a well-ordered manner, and the freedom of movement of individual molecules is very limited. When the cube dissolves, the freedom of movement of the molecules increases greatly and their random movements distribute them evenly throughout the available space. Subsequently, the tendency toward redistribution disappears and the entropy of the system is at its maximum. ΔS_{sys} increases

(b) The sugar molecules randomly distributed in a solution can only return to an organized state if the entropy of the environment increases, for example if the ordered water molecules in the liquid phase lose their organization through evaporation. ΔS_{sys} decreases associated with the increase in ΔS_{env} .

Example:

- The digestion of complex sugars (starch and glycogen) in the digestive tract: starch and glycogen are organized forms of carbohydrates, which are broken down into D-glucose, a smaller molecule that contains less energy and is less organized.
- Glucose in the cell is oxidized into CO₂ and H₂O, smaller and simpler molecules dispersed in the environment.
- The folding of a polypeptide chain or the formation of a lipid bilayer in membranes leads to more ordered states (decrease in entropy), associated with an increase in the disorder of the surrounding water molecules.

Spontaneous processes occur in a direction that increases the disorder of the universe, i.e., the system and its environment.

3/ Gibbs free energy (ΔG)

The change in Gibbs free energy ΔG , or free enthalpy, is the portion of a system's energy that produces **useful work**. Also called chemical potential.

Under conditions of constant temperature and pressure, the relationship between the change in free energy (ΔG) of a reacting system, and the change in entropy (ΔS) is given by the following equation:

$$\Delta G = \Delta H - T \Delta S$$

Where ΔH is the change in enthalpy (heat) and T is the absolute temperature in kelvins.

NB: Changes in enthalpy are due to energy exchanges, part of which is useful (G) and part of which is not (S).

In other words, Gibbs free energy (ΔG) allows us to evaluate the part of the system's potential energy (its enthalpy, ΔH) that is not dissipated as heat (its entropy, ΔS).

$$\Delta S_{\text{system}} + \Delta S_{\text{environment}} = \Delta S_{\text{universe}} > 0$$

$$\Delta S_{\text{environment}} = - \Delta H_{\text{system}} / T$$

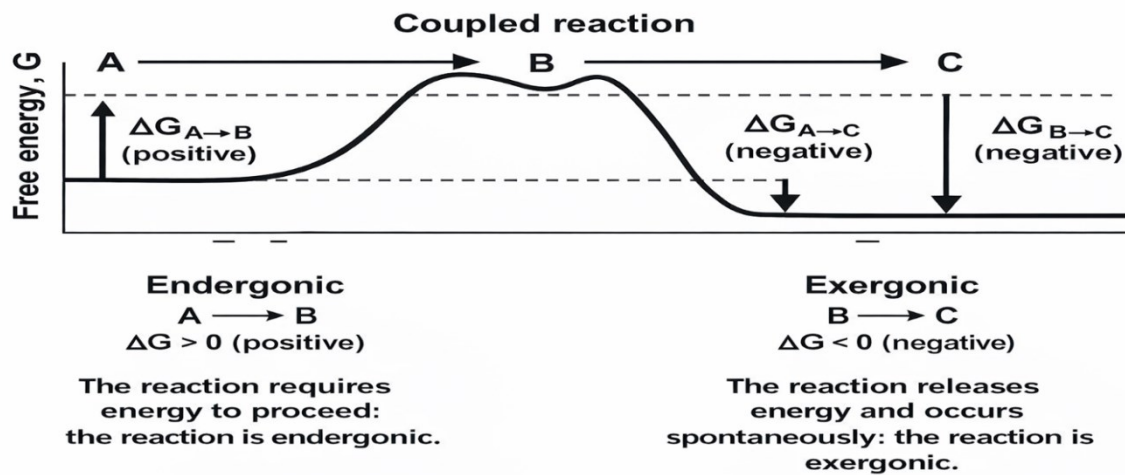
$$\Delta S_{\text{system}} - \Delta H_{\text{system}} / T = \Delta S_{\text{universe}}$$

$$T \Delta S_{\text{system}} - \Delta H_{\text{system}} = T \Delta S_{\text{universe}}$$

$$- T \Delta S_{\text{universe}} = - T \Delta S_{\text{system}} + \Delta H_{\text{system}}$$

Thus, the change in free energy of a reaction, ΔG depends both on the change in internal energy and on the change in entropy of the system. The ΔG is a valuable criterion in determining whether a reaction can occur spontaneously. Thus,

- If $\Delta G < 0$ in this case $\Delta H < T \Delta S$; the reaction will occur spontaneously and release energy toward environment (it is therefore exergonic). If, in addition, ΔG is of great magnitude, the reaction goes virtually to completion and is essentially irreversible.
- If $\Delta G > 0$, then $\Delta H > T \Delta S$ the reaction proceeds only if free energy can be gained, i.e., it is endergonic. If, in addition, ΔG is of high magnitude, the system is stable with little or no tendency for a reaction to occur.
- If $\Delta G = 0$, the system is in equilibrium. and no net change takes place.



4/ Relationship between ΔG and the equilibrium constant of a reaction K_{eq}

Measuring the free energy ΔG of a system requires defining a standard state or reference state. This standard state is the state in which an element or compound is most stable at ordinary temperature and pressure.

The Gibbs free energy of a system in its standard state is designated by the symbol: ΔG° .

Considering the reaction: $A + B \rightleftharpoons C + D$

- For chemists, the standard conditions of a system are:
 - a pressure P of 1 atmosphere (101.3 kPa)
 - a temperature of 25°C, or 298 degrees Kelvin
 - a solute concentration of 1 M $[A] = [B] = [C] = [D]$
 - consequently: pH = 0 (since: $pH = -\log [H^+]$ and $[H^+] = 1 \text{ M}$)

Standard conditions are different in biochemistry, since all reactions in cells take place in a highly diluted aqueous medium at pH 7.

- The standard conditions for biochemists are therefore:
 - a pH of 7 and therefore a concentration $[H^+] = 10^{-7} \text{ M}$
 - a water concentration (55.5 M) that is considered constant and does not appear in the expression of equilibrium constants

Consequently, the standard Gibbs free energy of a biological system is designated by the symbol: ΔG° .

❖ Relationship between ΔG° and ΔG (actual)

Under real conditions of cellular life, the temperature is not 25°C, but close to 37°C, and the initial concentrations of products and reactants are not 1M, but unequal and much lower than 1M.

For a given reaction: $aA + bB \rightleftharpoons cC + dD$

The following relationship can be used to calculate the actual non-standard free energy change

$$\Delta G: \Delta G = \Delta G^\circ + R.T \ln [C]^c . [D]^d / [A]^a . [B]^b$$

$$= \Delta G^\circ + R.T \ln K \text{ in KJ.mol}^{-1}$$

- ✓ Where $[A]$, $[B]$, $[C]$, and $[D]$ are the initial concentrations of the reactants.
- ✓ T: absolute temperature in Kelvin ($T + 273K$)
- ✓ R: ideal gas constant, $R = 8.314 \text{ J.mol}^{-1}\text{K}^{-1} = 1.99 \text{ cal.mol}^{-1}\text{K}^{-1}$

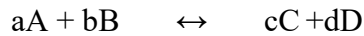
✓ $\ln K$: natural logarithm of the reaction constant

The change in free energy ΔG is not a constant; it depends on:

- The nature of the reacting substances, represented by the term ΔG° .
- The reaction conditions: temperature and initial concentrations of the reactants.

This relationship can be expressed as $\Delta G = Cte + \text{correction factor}$.

❖ Fundamental relationship between ΔG° and K



At equilibrium, $\Delta G = 0$

$$\Delta G = \Delta G^\circ + RT \ln [C]^c * [D]^d / [A]^a * [B]^b ,$$

$$0 = \Delta G^\circ + RT \ln [C]^c * [D]^d / [A]^a * [B]^b ,$$

$$\Delta G^\circ = - RT \ln [C]^c * [D]^d / [A]^a * [B]^b$$

$$\Delta G^\circ = - RT \ln K_{eq}$$

$K = 1, \Delta G^\circ = 0$ State of equilibrium, no displacement $[C]_{eq}[D]_{eq} = [A]_{eq}[B]_{eq}$

$K < 1, \Delta G^\circ > 0$ Reaction impossible in the direction of formation of C and D, endergonic reaction, requires external energy input. $[C]_{eq}[D]_{eq} < [A]_{eq}[B]_{eq}$. It took place in the reverse direction.

$K > 1, \Delta G^\circ < 0$ Spontaneous reaction in the direction of formation of C and D. It is exergonic.
 $[C]_{eq}[D]_{eq} > [A]_{eq}[B]_{eq}$

Note that a reaction at equilibrium does not mean equal concentrations of metabolites.

The **free energy change** ΔG of a reaction therefore depends on the standard free energy and concentration of each metabolite. This concept determines the spontaneity and reversibility of the reaction. If $[A]$ is close to the value $[A]_{eq}$, then the reaction is spontaneous and easily reversible, whereas if $[A]$ differs too much from $[A]_{eq}$, then it will be irreversible.

ΔG therefore measures the distance between the system at the concentrations considered and those at equilibrium.

IV. ADDITIVITY OF ΔG : REACTION COUPLING MECHANISM

Many metabolic reactions have an unfavorable ΔG° , which would not spontaneously allow the synthesis of compounds that the body needs.

These reactions can occur through energy coupling (chemical association) that combines an endergonic reaction with an exergonic reaction.

In other words, coupling a highly exergonic reaction with a reaction that is less endergonic than the first results in an overall reaction whose free energy change ΔG° is sufficiently negative for the overall reaction to be spontaneous.

Conditions:

- The sum of the ΔG° values of the two reactions must be negative.
- The two reactions must share a common intermediate.

Exergonic reactions are grouped under the term catabolism: degradation or oxidation of molecules used as fuel; and endergonic reactions under the term anabolism: synthesis of substances.

The combined catabolic and anabolic processes constitute metabolism.

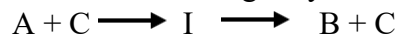
Reaction 1: Glucose + Pi $\longrightarrow$ G6P $\Delta G^{\circ} = + 13 \text{ kcal/mol}$ (endergonic)

Reaction 2: ATP $\longrightarrow$ ADP + Pi $\Delta G^{\circ} = - 7 \text{ Kcal/mol}$ (exergonic)

Overall reaction: Glucose + ATP $\longrightarrow$ G6P + ADP $\Delta G^{\circ} = - 4 \text{ Kcal/mol}$ Possible reaction

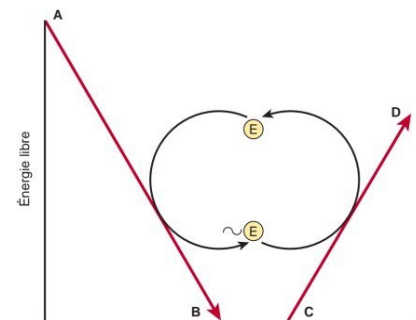
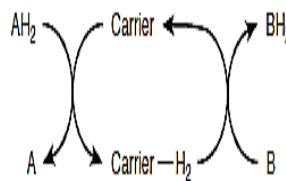
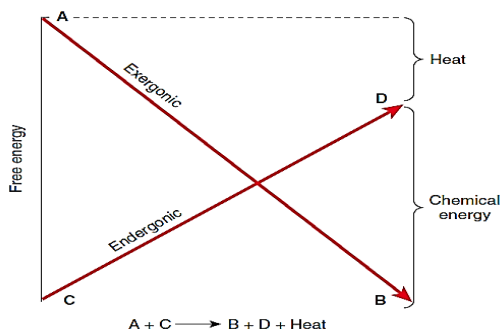
Different coupling methods are possible in cells:

- 1- Presence of a common obligatory intermediate involved in both reactions, i.e.:



E.g.: dehydrogenation reactions that are coupled to hydrogenations by an intermediate transporter.

- 2- The synthesis of a substance with high potential energy (E) in the exergonic reaction and the incorporation of this new substance into the endergonic reaction. (E) can serve as an energy transducer for a wide range of exergonic and endergonic reactions. E.g.: ATP



V. FREE ACTIVATION ENERGY $\Delta G^{\#}$

During a reaction, energy exchanges with the surrounding environment take place:

- Some bonds in the substrate are broken by absorbing energy.
- And the bonds of the product are formed by releasing energy.

Free activation energy $\Delta G^{\#}$:

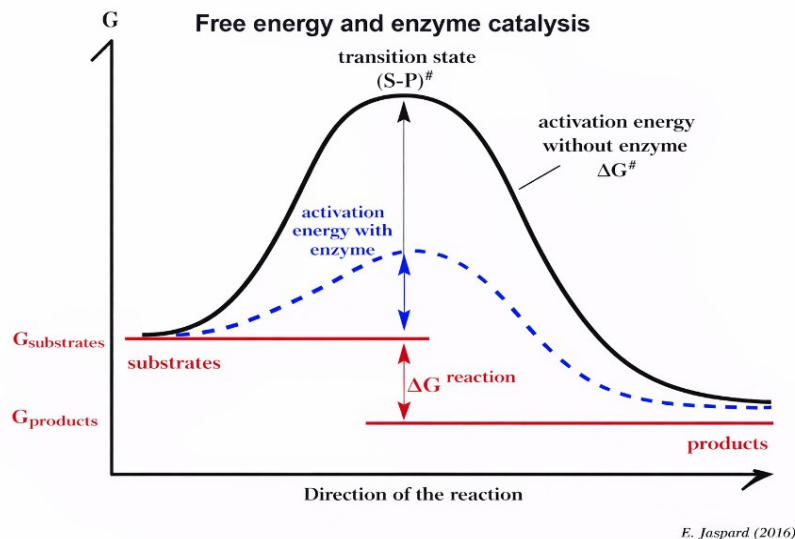
This is the energy required to break the bonds of the reactants and is therefore necessary for the reaction to take place.

The absorption of this energy causes the substrates to move from an initial state to a transition state. The speed of the reaction is inversely proportional to $\Delta G^{\#}$.

Transition state:

This is a higher energy state characterized by distortion of the angles and lengths of the chemical bonds of the substrates, which are closely linked to each other. This state is unstable and spontaneously evolves towards a lower energy state, the product of the reaction.

- It should be noted that even in the case of an exergonic reaction, the substrates must cross the activation barrier.
- This barrier is essential for life: without it, macromolecules (proteins, nucleic acids, etc.) with high energy potential would spontaneously decompose.

**Role of enzymes:**

An enzyme increases the reaction rate by lowering the activation energy: at the same temperature, substrates cross the activation barrier more easily and therefore more frequently.

Enzymes do not alter the Gibbs free energy change of the reaction: the dissociation equilibrium constant of a reaction is not altered.

VI. CONCEPT OF "ENERGY-RICH" OR HIGH-ENERGY POTENTIAL COMPOUNDS

These are molecules with bonds whose hydrolysis releases a lot of energy. They have a high hydrolysis potential, with ΔG° hydrolysis ≤ -30 kJ/mole being the threshold.

There are five types of energy-rich bonds

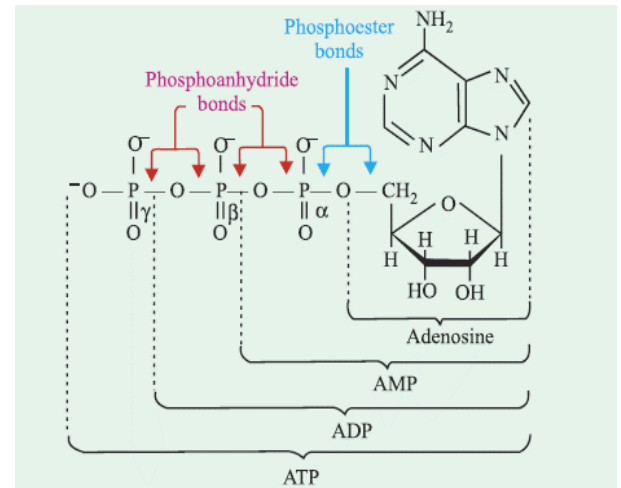
- 1- Phosphoric anhydride
- 2- Mixed phosphoric anhydride
- 3- Enol phosphate
- 4- Thioester (Acetyl CoA)
- 5- Guanidine phosphate - Phosphocreatine

The hydrolysis of these bonds is accompanied by the release of a large amount of energy used in energy coupling.

1- ATP: Adenosine triphosphate

ATP is a nucleotide consisting of:

- Purine nitrogen base: Adenine
- D-ribose
- 3 phosphate groups: α , β , and γ
- 2 energy-rich bonds: The bond between the first (α) and second (β) phosphoryl groups and between the second (β) and third (γ) groups is a phosphoanhydride bond.



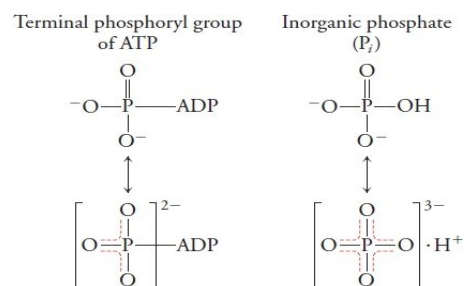
The active form of ATP is an ATP-Mg^{2+} complex. ATP is doubly energy-rich:

- In certain reactions, it loses a phosphate and becomes ADP.
- In some cases, 2 phosphates become AMP.

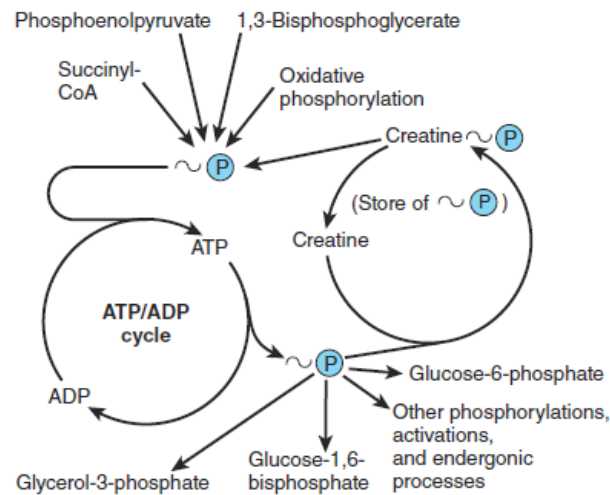


This is explained by the fact that the hydrolysis products of ATP are much more stable than ATP.

- The hydrolysis products of ATP are more stable than the reactants. At physiological pH, ATP has three to four negative charges (its pK is close to 7), and the anionic groups repel each other. In the products ADP and P_i , the separation of charges attenuates part of this unfavorable electrostatic repulsion.
- A compound with a phosphoanhydride bond undergoes less resonance stabilization than its hydrolysis products. Resonance stabilization reflects the degree of electron delocalization in a molecule. There are fewer equivalent ways to arrange the bonds of the terminal phosphoryl group of ATP than in free P_i .



ATP plays a central role in the transfer of free energy from exergonic reactions to endergonic reactions. ATP is an intermediate between catabolism and anabolism.



ATP synthesis can occur in two ways:

- Substrate-level phosphorylation.

The other mode is the synthesis of ATP by ATP synthase following a series of reactions that take place in the mitochondria: the respiratory chain.

1- Phosphoenolpyruvate

The most energetic phosphate bond in the cell is that of phosphoenolpyruvate (PEP), an intermediate in glycolysis: $\Delta G^{\circ} \approx -14.8$ kcal/mol.

In glycolysis, pyruvate kinase catalyzes the transfer of the phosphoryl group from PEP to ADP to form ATP and pyruvate.

Despite the formation of an ATP molecule, the reaction is highly exergonic: $\Delta G^{\circ} \approx -14.8 - (-7.3) \approx -7.5$ kcal/mol and represents one of the main sources of ATP in the cell.

2- Phosphocreatine

A working muscle cell, for example, renews its entire ATP supply about once per minute. This represents 10 million ATP molecules used and regenerated per second per cell.

At the start of muscular effort, creatine kinase catalyzes the supply of ATP by transferring the activated phosphoryl group from phosphocreatine.

Phosphocreatine and phosphoarginine are phosphagens. Their bonds are guanidine phosphate and not phosphoanhydrides.

3- Coenzyme A

Coenzyme A, or CoA or CoASH, is the molecule that enables acyl group ($R-C=O$) transfer reactions, for example, during fatty acid catabolism.

These groups are linked to coenzyme A by thioester bonds, which are high-energy bonds ($\Delta G^{\circ} \approx -9$ kcal/mol).

Coenzyme A is a derivative of pantothenic acid, a vitamin belonging to the B vitamin family.

Compound	$\Delta G^{\circ'}$	
	kJ/mol	kcal/mol
Phosphoenolpyruvate	-61.9	-14.8
Carbamoyl phosphate	-51.4	-12.3
1,3-Bisphosphoglycerate (to 3-phosphoglycerate)	-49.3	-11.8
Creatine phosphate	-43.1	-10.3
ATP $\rightarrow$ AMP + PP _i	-32.2	-7.7
ATP $\rightarrow$ ADP + P _i	-30.5	-7.3
Glucose-1-phosphate	-20.9	-5.0
PP _i	-19.2	-4.6
Fructose-6-phosphate	-15.9	-3.8
Glucose-6-phosphate	-13.8	-3.3
Glycerol-3-phosphate	-9.2	-2.2